EXPERIMENT 21

Implement the morphological opening operation using OpenCV.

Aim

Implement the morphological opening operation using OpenCV.

Algorithm

• Import OpenCV and NumPy.

• Read the input image.

• Convert it to grayscale and binary form.

• Apply opening using a structuring kernel.

• Save the processed image.

Code

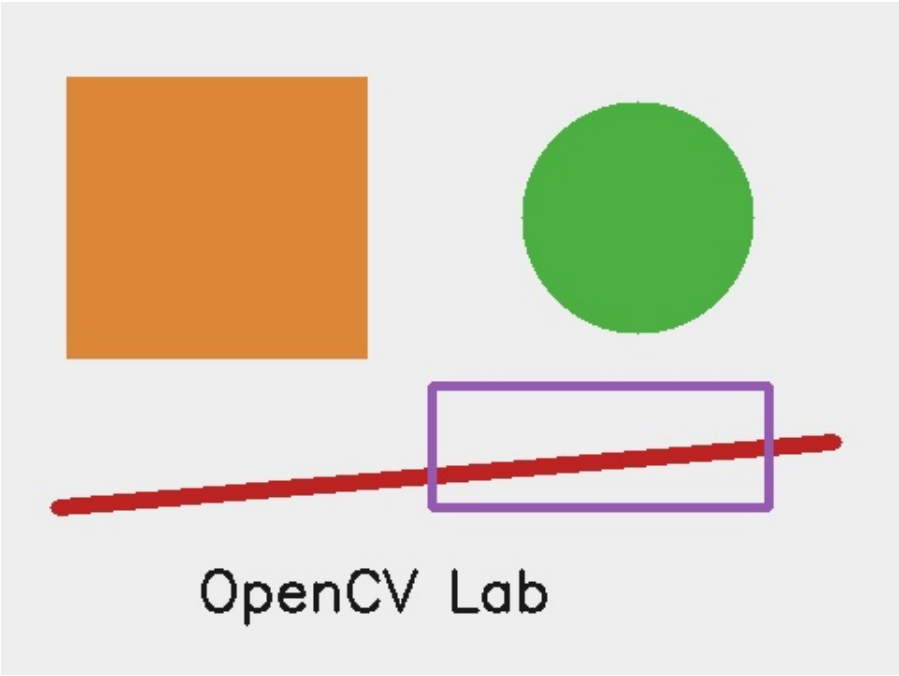
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
kernel = np.ones((5, 5), np.uint8)
result = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
cv2.imwrite("output_images/experiment21_opening.png", result)
print("Saved output_images/experiment21_opening.png")
```

Input Image:

Output Image:

Note: Morphological operations use a thresholded binary version of the sample image.

Input Image



Output Image

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